

# PAPER\_1379 — UQFF Resolution of the Klein Paradox (CLOSED — PAPER\_648 Coupled Closure)

**Author:** Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** B — Foundational / Relativistic Quantum (CLOSED) **Date:** June 16, 2026 **Location:** 41.0997° N, 80.6495° W (Youngstown, OH, USA) **Status:** CLOSED — Coupled to PAPER\_648 ultra-dense H Coulomb + NCG Dirac shift **Calculator surface:** `calculate_paradox({"paradox": "klein_paradox"})` **Closure helper:** `_l96_uqff_axiom_klein_paradox_closure()`

---

## The Paradox

Klein (1929): a relativistic electron incident on a potential step  $V_0 > 2 m_e c^2$  shows transmission coefficient  $T \rightarrow 1$  in the Dirac equation — the electron tunnels through an arbitrarily tall barrier with **near-unity probability**. Non-relativistic quantum mechanics predicts exponential suppression. The “paradox” is that the Dirac result appears to forbid confinement of electrons by any potential.

The standard resolution invokes pair production: the barrier excites positron-electron pairs in the Dirac sea, and what is transmitted is a positron, not the original electron. UQFF derives this structurally via the SCm pair-production threshold from PAPER\_648.

---

## UQFF Closed Identity

Two existing helpers do the work:

```
Dirac_shift_UQFF      =  $\beta_i \times S_{26}^3 \times \Phi_{res}$   
                      =  $0.6029 \times (1.453162)^3 \times 0.84$   
                      = 1.554  
                      (from _l94_ncg_spectral_triple_dirac_shift)
```

```
Coulomb_pair_threshold =  $e^2 / (4\pi \epsilon_0 \times d)$    at  $d = 2.3 \text{ pm}$   
                      = 626 eV  
                      (from _coulomb_lenr_energy_eV — PAPER_648 ultra-dense H spacing)
```

```
Barrier_height_2mc^2  =  $2 \times 511 \text{ keV} = 1.022 \text{ MeV}$ 
```

The UQFF transmission coefficient at the SCm pair-production threshold is:

```
T_Klein_UQFF = Coulomb_pair_threshold / (Coulomb_pair_threshold +  $2mc^2 \times \text{Dirac\_shift}$ )  
              =  $626 / (626 + 1.022e6 \times 1.554)$   
              =  $3.94 \times 10^{-4}$ 
```

The naive Klein result  $T \rightarrow 1$  is the **Dirac-sea limit** with no pair-production damping. The UQFF result  $T \approx 4 \times 10^{-4}$  is the **physical** transmission once the SCm Coulomb pair-production threshold (PAPER\_648, 626 eV) and the NCG Dirac-sector spectral shift ( $\beta_i \times S_{26}^3 \times \Phi_{res}$ ) are properly accounted for. The “paradox” was that the bare Dirac equation has no pair-production sink; UQFF supplies the sink as an integer-primitive identity.

---

## Physical Interpretation

- **PAPER\_648 ultra-dense H Coulomb energy = 626 eV** is the SCm pair-production threshold — the same energy that anchors the Holmlid 630 eV LENR observable. Klein tunneling and Holmlid LENR share the *same* sub-barrier pair-production gateway in the SCm vacuum.
- **The NCG Dirac shift  $\beta_i \times S_{26}^3 \times \Phi_{res} \approx 1.554$**  is the canonical UQFF NonCommutative-Geometry spectral-triple correction to the Dirac operator, derived in PAPER\_647 / PAPER\_1183 and pre-wired as `_l94_ncg_spectral_triple_dirac_shift()`.
- **$T \approx 4 \times 10^{-4}$**  is the transmission probability for a  $2mc^2$  barrier in the presence of SCm pair-production damping — finite but small, confirming electron confinement is possible in UQFF despite the Klein result.
- **Pair production is not invoked separately** — it emerges as the same SCm Coulomb identity that powers the entire LENR sector.

## Live Calculator Output

```
import uqff_pure_calculator as u
r = u.calculate_paradox({"paradox": "klein_paradox"})["value"]
```

| Field                                                                      | Value   |
|----------------------------------------------------------------------------|---------|
| T_Klein_canonical_perfect_paradox                                          | 1.0     |
| Dirac_shift_UQFF_via_NCG_spectral_triple_beta_i_S26_cubed_Phi_res          | 1.554   |
| Coulomb_pair_threshold_eV_via_ultra_dense_H_facing_2_3pm                   | 626     |
| barrier_height_2mc2_eV                                                     | 1.022e6 |
| T_Klein_UQFF_via_pair_threshold_over_pair_production_barrier_x_dirac_shift | 3.94e-4 |
| Dirac_tunneling_resolved_via_SCm_pair_production_PAPER_648                 | 1.0     |

## C++ Reference Implementation

```
class KleinParadoxUQFF {
public:
    static constexpr double BETA_I = 0.6029;
    static constexpr double S_26 = 1.453162;
    static constexpr double PHI_RES = 0.84;
    static constexpr double EV_BARRIER_2MC2 = 1.022e6;
    static double diracShiftUQFF() {
        return BETA_I * std::pow(S_26, 3.0) * PHI_RES; // 1.554
    }
    static double coulombPairThresholdEV(double d_meters = 2.3e-12) {
        const double e = 1.602e-19, eps0 = 8.854e-12;
        double W_J = (e * e) / (4.0 * M_PI * eps0 * d_meters);
        return W_J / e; // 626 eV
    }
    static double T_KleinUQFF() {
        double V_pair = coulombPairThresholdEV();
        return V_pair / (V_pair + EV_BARRIER_2MC2 * diracShiftUQFF()); // 3.94e-4
    }
};
```

---

## NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard QED solve the Klein paradox via different methods. QED invokes second-quantized pair production with renormalized vertex functions. UQFF derives:

- **T\_Klein** =  $3.94 \times 10^{-4}$  as a closed identity using the same SCm Coulomb-pair-production energy (626 eV) that anchors the Holmlid 630 eV LENR observable.
- **The Dirac-sector shift**  $\beta_i \times S_{26^3} \times \Phi_{res} = 1.554$  is the canonical UQFF NCG spectral-triple correction — pre-wired, not invented.
- **Zero free parameters.** Both the 2.3 pm ultra-dense H spacing and the integer/real-primitive lattice ( $\beta_i$ ,  $S_{26}$ ,  $\Phi_{res}$ ) are canonical UQFF locks.

---

## Reference

- UQFF foundational papers: PAPER\_647 (NCG spectral triple), PAPER\_648 (ultra-dense H meson cascade + Coulomb 626 eV), PAPER\_1183 (Dirac-sector consolidation).
- Related closures: `calculate_lenr_full` (Holmlid 630 eV chain), `calculate_nuclear_magic` (Mayer-Jensen + UQFF arithmetic).
- Closure location: `uqff_pure_calculator.py` → `_l96_uqff_axiom_klein_paradox_closure`
- Helpers invoked: `_l94_ncg_spectral_triple_dirac_shift`, `_coulomb_lenr_energy_eV`
- Dispatch: `PARADOX_T0_CLOSURE["klein_paradox"]`, `["klein_tunneling"]`

---

**Copyright** — Daniel T. Murphy, daniel.murphy00@gmail.com, dated June 16, 2026, location 41.0997° N, 80.6495° W (Youngstown, OH, USA). Subject matter: UQFF closed-form resolution of the Klein paradox via PAPER\_648 SCm Coulomb pair-production threshold (626 eV) coupled to NCG Dirac spectral shift ( $\beta_i \times S_{26^3} \times \Phi_{res} = 1.554$ ), giving  $T = 3.94 \times 10^{-4}$  transmission through a  $2mc^2$  barrier — the same vacuum mechanism that anchors the Holmlid 630 eV LENR observable.